

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Manoj Kumar K	Reg. No.:	192521041
Section / Batch:	Information Technology(IT)	Date:	24/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Q1.An inventory system searches for item IDs in an unsorted dataset using Linear Search.

a) Explain how Linear Search is applied. b) Analyze performance issues. c) Suggest improvements and justify.

Q2.A smart grid system collects power consumption readings that arrive in random order and must be sorted periodically for analysis. a) Analyze how Selection Sort and Bubble Sort perform on this dataset. b) Compare their efficiency in terms of comparisons and swaps. c) Recommend the most suitable algorithm with justification.

Code of Conduct

I Manoj Kumar K / 192521041 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO2- ASSESSMENT TOOL 2

1. An inventory system searches for item IDs in an unsorted dataset using Linear Search.

a) Application of Linear Search

In an inventory system, item IDs are stored in an unsorted dataset. Since the data is not arranged in any order, Linear Search can be used to find a specific item ID. The algorithm starts checking from the first item and compares each item ID with the required search ID. It continues sequentially until the item is found or the end of the dataset is reached.

For example, if the inventory contains item IDs **105, 203, 150, 320, and 450**, and the system searches for item ID **320**, Linear Search checks each item one by one: **105 → 203 → 150 → 320**. When the matching item is found, the search stops and returns its position.

b) Performance Issues

The main performance issue with Linear Search is that it may require checking every item in the dataset. If the required item is located at the end or is not present, the algorithm performs the maximum number of comparisons.

The time complexity of Linear Search is:

- **Best Case:** $O(1)$ – Item is found at the first position.
- **Average Case:** $O(n)$ – Item is found somewhere in the middle.
- **Worst Case:** $O(n)$ – Item is at the end or not found.

For a large inventory system containing thousands of item IDs, Linear Search can become slow and inefficient because each search may require multiple comparisons.

c) Suggested Improvements and Justification

Performance can be improved by using a more efficient search technique. If the inventory data can be sorted, **Binary Search** can be used. Binary Search reduces the search area by half after every comparison and has a time complexity of **$O(\log n)$** .

Another improvement is to use a **Hash Table**, where item IDs can be stored using a hash function. This provides an average search time of **$O(1)$** .

Therefore, for a large inventory system, a **Hash Table is the most suitable improvement** because item IDs can be searched quickly without checking every item sequentially. If maintaining sorted data is practical, Binary Search is also a suitable alternative.

2. A smart grid system collects power consumption readings that arrive in random order and must be sorted periodically for analysis.

)Selection Sort and Bubble Sort

In a smart grid system, power consumption readings arrive in random order. These readings need to be sorted for analysis.

Selection Sort finds the smallest value from the unsorted data and places it in the correct position. It repeats this process until all readings are sorted.

Bubble Sort compares adjacent values and swaps them if they are in the wrong order. It repeats the process until the data is sorted.

Both algorithms have a time complexity of **$O(n^2)$** for average and worst cases.

Comparison of Comparisons and Swaps

Feature	Selection Sort	Bubble Sort
Comparisons	$O(n^2)$	$O(n^2)$
Swaps	Less	More
Performance	Better when swaps are costly	More data movement

Selection Sort generally performs fewer swaps than Bubble Sort. Bubble Sort may repeatedly swap adjacent elements many times.

Recommendation

Selection Sort is recommended for this dataset because it performs fewer swaps than Bubble Sort. This makes it more efficient when data movement is costly.

However, if the smart grid contains a very large number of readings, **Merge Sort or Quick Sort** would be better because their average time complexity is **$O(n \log n)$** .

Conclusion: For comparing only Selection Sort and Bubble Sort, **Selection Sort is the better choice** because it uses fewer swaps.